

Unit 6, Lesson 12: Solving Problems Involving Ratios

Benchmark

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

Learning Target

- ☐ I can solve problems involving ratios, rates, and unit rates.

6.12.1 Warm-Up: Bell Work

1. Mai has to fill and seal 60 packages of homemade trail mix to sell at the upcoming farmer's market. She filled and sealed 25 packages in 32 minutes. Mai wonders, "If I keep working at this rate, will I be able to get all of these done in less than 1 hour?"

Explain whether Mai will be able to get all 60 packages filled in less than 1 hour working at this rate.

6.12.2 Activity: Station 1: Is It a Deal?

Your teacher will give you a set of cards showing different offers.

1. Find card A and work with your partner to decide whether the offer on card A is a good deal. Explain or show your reasoning in the space provided.
2. Next, split cards B–E so you and your partner each have two cards.

a. Decide individually if your two cards are good deals. Explain your reasoning for your two cards in the table.

Is the new deal a good deal?			
Card B	Card C	Card D	Card E

- b. For each of your cards, explain to your partner if you think it is a good deal and why.
- c. Listen to your partner’s explanations for their cards. If you disagree, explain your thinking. Write your partner’s explanation for their cards in the table above, after resolving any differences.

- d. Revise any decisions about your cards based on the feedback from your partner.
3. When you and your partner are in agreement about cards B–E, place all the cards you think are a good deal in one stack, and all the cards you think are a bad deal in another stack.
4. Another potential deal is shown.

Fruit Snacks



Original: 12 for \$9.12
New Deal: 9 for _____

- a. What is the price for 9 packs of fruit snacks at the same unit price as the original?

- b. Complete the statement.

Any price less than \$_____ for 9 packs of fruit snacks would be considered a good deal because the cost per pack would be less than \$_____.

6.12.3 Activity: Station 2: Cooking with a Tablespoon

Diego is trying to follow a recipe, but he cannot find any measuring cups. He only has a tablespoon.

In the cookbook, it says that 1 cup equals 16 tablespoons.

1. How could Diego use the tablespoon to measure out these ingredients?
 - a. $\frac{1}{2}$ cup almonds
 - b. $1\frac{1}{4}$ cup of oatmeal
 - c. $2\frac{3}{4}$ cup of flour
2. Diego also adds the following ingredients. How many cups of each did he use?
 - a. 28 tablespoons of sugar
 - b. 6 tablespoons of cocoa powder

6.12.4 Activity: Station 3: Turbo Texting

1. Jonah was with his friend Erin after school and he needed to text his mom. Erin watched Jonah text and commented on how slow he was.

They decided to have a text race to determine who was faster. The image below shows what each person texted in the race.



- a. Discuss any initial observations with your partner.
- b. Erin was able to send her text slightly faster than Jonah. Explain if you think this means she is the faster texter.

- c. Describe the relationship you notice between the number of characters in a text versus how long it might take to type it.
- d. What additional information would be helpful to determine who is the faster texter?

2. The image below gives additional information about each person's text.

I am the fastest texter!!! You are so slow! You will never beat me!

67 characters
7.4 seconds

Think so? You're about to lose this! It's on!!

46 characters
6.5 seconds

Work with your partner or small group to determine who is the faster texter, using your preferred strategy. Show your work.

3. Jonah and Erin decided to send one last text to see who could send it out faster. The image shows the text they each sent.

So let's settle this once and for all. I am a way faster texter and there is nothing you can do about it. I am the GOAT and you'll never catch me! I won!!

154 characters

So let's settle this once and for all. I am a way faster texter and there is nothing you can do about it. I am the GOAT and you'll never catch me! I won!!

If both people sent the message above at the same rate as before, predict about how many seconds it took each person to text the message to the nearest tenth of a second. Show your work.

Person	Predicted Time
Jonah	
Erin	

6.12.5 Wrap-Up: Reflection

1. Complete the learning period based on today's lesson.

One question I would like to have
answered ...

One mistake I learned from today ...

One thing I am not sure about ... YET!

One thing I am confident I know is ...